Kernel

$K(x_i, x_j) \longrightarrow$ Similarity b/w x_i and x_j

$$\underbrace{K(x_i, x_j)}_{\text{Simply computable expression}} = \phi(x_i)^T \phi(x_i)$$

Simply computable expression

e.g. $e^{-a\|x_i - x_j\|^2}$

output in scalar

$\phi(x_i)$ → theoretical vector features computed using x_i, can have inf. dimension

→ Ensure that given matrix is P.S.D.

↳ positive semi-definite, eigen value ≥ 0

$$\begin{bmatrix} & | & \\ — & K(x_i, x_j) & — \\ & | & \end{bmatrix}$$

and $K(x_i, x_j) = K(x_j, x_i)$

Kernel trick

1) Take an objective f^n of x and try to change it prime so that it become of a f^n of (x_i^T, x_j) or $\phi(x_i)^T \phi(x_i)$ or $K(x_i, x_j)$

2) Replace (x^T, x_j) with $\phi(x_i)^T \phi(x_i)$

Primal SVM objective

$$L(w, b) = \frac{1}{2}\|W\|_2^2 - \sum_i a_i \left[(y_i (w^T x + b) - 1 \right]$$

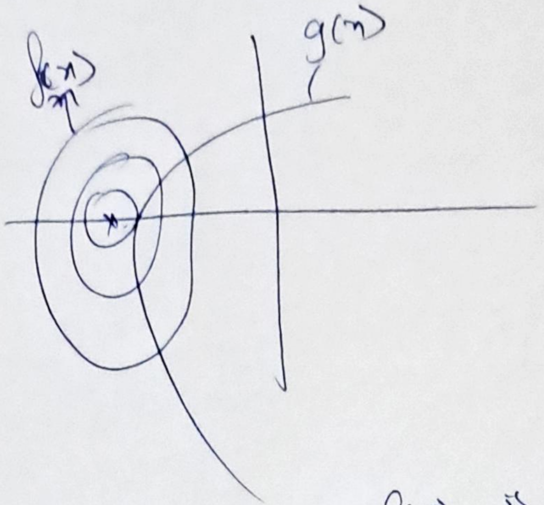
~~Background~~

(ii) ~~Primal SVM algo~~

$\cancel{L(w,b) = \frac{1}{2}\|w\|}$

Background: $L(x, \lambda) = f(x) + \lambda g(x)$

$\lambda \geq 0$



$\nabla f(x)$ is opp dir. to $\nabla g(x)$

case 1 $g(x) = 0$; $\lambda > 0$

case 2 $g(x) > 0$; $\lambda = 0$ {$\nabla g(x)$ does not matter}

$L(w,b,a) = \frac{1}{2} w^T w - \sum_i a_i (t_i (w^T x_i + b) - 1)$; $a \geq 0$

$\frac{\partial L}{\partial w} = 0 \Rightarrow w = \sum_i a_i t_i x_i$

$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum a_i t_i = 0$

Dual : $L(a) = \sum_i a_i - \frac{1}{2} \sum_i \sum_j (a_i, a_j) t_i, t_j x_i^T x_j$

$L(w, b, a) = \frac{1}{2} w^T w - \sum a_i (t_i (w^T x_i + b))$

$(\sum_i a_i t_i x_i)^T (\sum_j a_j t_j x_j) = \sum_i \sum_j a_i a_j t_i t_j x_i^T x_j$

$y_i = w^T x + b$

$y_i = \sum_j a_j t_j x_j^T x_i + b$ {S → set of support vectors}

$b = \frac{1}{N_S} \cdot \sum_{i \in S} (t_i - \sum_{j \in S} a_j t_j x_i^T x_j)$